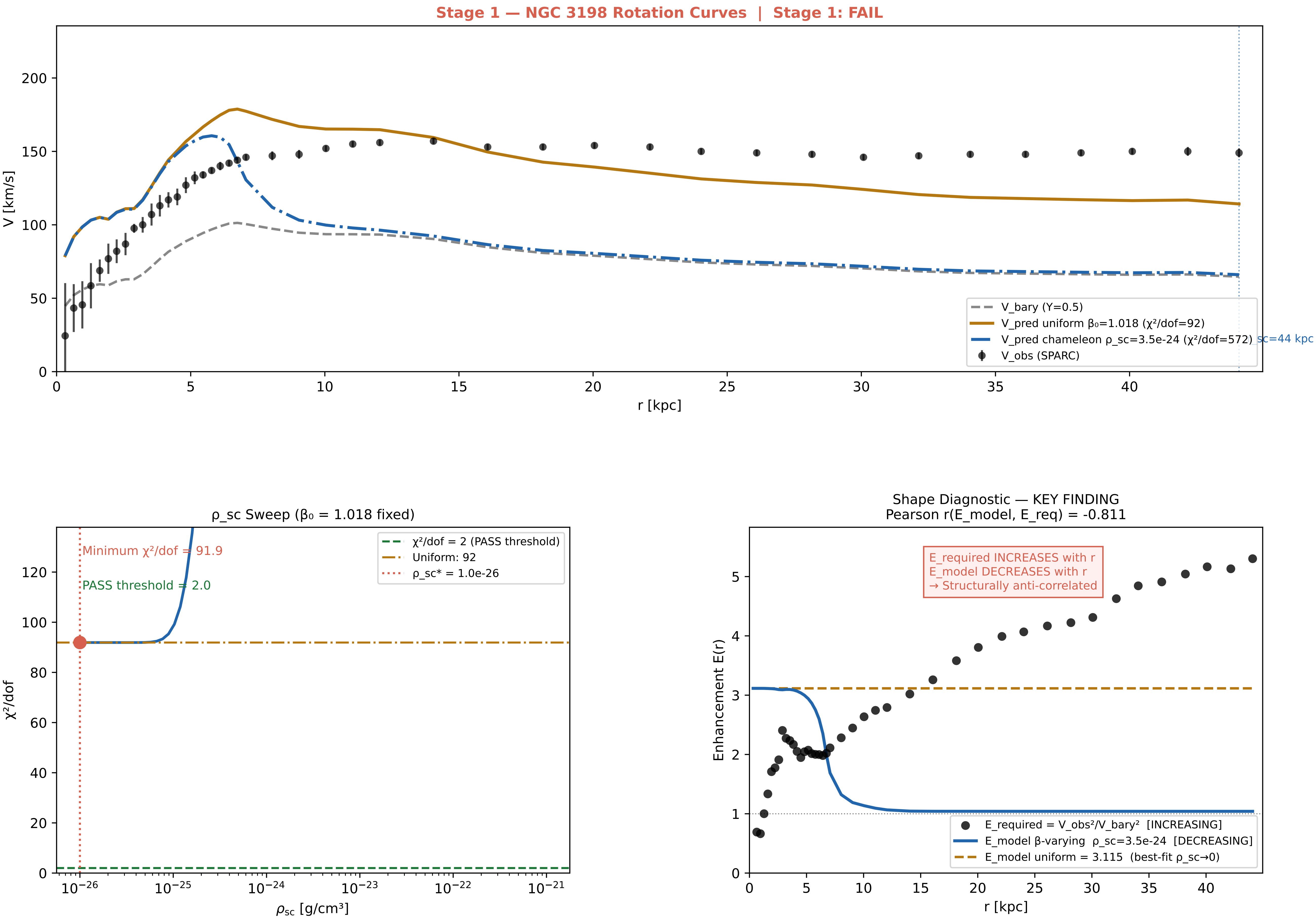
SIM152 — Paper V Option B: SPARC Rotation Curve Test
Stage 1 (FAIL) | Stage 2 (NOT REACHED)



SIM152 — Results Summary

Quantity	Result	Status
Stage 1: uniform β_0 = 1.018	χ^2/dof = 91.9	FAIL
Stage 1: radial $\beta(\rho_b)$, best ρ_{sc}*	χ^2/dof = 91.9 (ρ_{sc}* = 1.00e-26 g/cm³)	FAIL
Stage 1: free β_0 at ρ_{sc}*	β_0 = 0.9990 (drift 2% from 1.018) χ^2/dof = 91.4	FAIL
Screening transition radius	r_{sc} = 44.1 kpc (screened mass frac = 100%)	—
Shape correlation E_model vs E_required	Pearson r = -0.811 (anti-correlated)	STRUCTURAL MISMATCH
E_required(r) trend	INCREASING: 1.8 at 2 kpc → 5.2 at 40 kpc	—
E_model(r) trend	DECREASING: 3.1 at 2 kpc → 1.04 at 40 kpc	INVERTED
Physical diagnosis	$\beta \propto \tanh^2(\rho_b/\rho_{sc})$ large where ρ_b large (inner disk)	—
	but flat curves require more enhancement where ρ_b SMALL (outer disk)	—
	No ρ_{sc} can reconcile this — structural, not parameter-tuning issue	
Stage 1 VERDICT	FAIL	FAIL
Stage 2	NOT REACHED	NOT REACHED
Option B viability	ARCHITECTURE INVERTED — $\beta(\rho)$ must DECREASE with density for flat curves	NOT VIABLE as specified